

Detailed Rewritten Thinking Log for the Two-Minus Benchmark

Codex run: `codex_55_xhigh`

June 2026

1 Task and Constraints

The task asked for a closed-form analytic formula for the tree-level amplitude A_n in the two-minus sector

$$\sigma = (-1, -1, +1, +1, \dots, +1),$$

valid for all $n \geq 4$. The prompt stated that the answer should be a single global rational function of the frequencies, with no chamber decomposition, no absolute values, and no piecewise conditions.

The prompt also imposed a file-reading constraint: only `prompt.md` and `OnShellBG.m` were allowed as pre-existing inputs. The directory already contained a sibling solution folder, but that folder was treated as off-limits. All generated scripts and outputs were placed in

`/home/zihanz/waterhedron_benchmark_blind/case_1/codex_55_xhigh.`

2 Initial Orientation

Directory and file checks

The first operational step was to locate the case files and confirm the output location. The command

```
pwd && rg --files waterhedron_benchmark_blind/case_1 \
  waterhedron_benchmark_blind_case_1 2>/dev/null | head -200
```

showed that the relevant files were in `/home/zihanz/waterhedron_benchmark_blind/case_1`, including `prompt.md` and `OnShellBG.m`. A search for `waterhedron_benchmark_blind_case_1` did not find a separate case directory, so the requested output folder was interpreted as a subfolder of `case_1`.

Understanding the supplied code

The supplied Wolfram file contained:

- exact kernels `EKernel` and `FKernel`;

- the symmetrized `Vertex`;
- the propagator `Propagator`;
- set partitions for the Berends–Giele recursion;
- `BGCurrent` and `BGAmplitude`;
- `MakeKinematics`, which solves the conservation constraints.

The relevant kinematics are

$$\sum_i \omega_i = 0, \quad \sum_i \sigma_i \omega_i^2 = 0.$$

The code sets $k_i = \sigma_i \omega_i^2 / g$, and in all tests here $g = 1$.

3 Runtime Setup and Early Blocker

Checking tools

The environment had both WolframScript and Python:

```
/opt/sns/bin64/wolframscript
/usr/bin/python3
Python 3.6.8
```

Running the whole stock Wolfram file was not useful because its built-in tests entered a slow symbolic $n = 8$ example. That run was interrupted. A later process table check showed a lingering `WolframKernel`, which was killed.

For a period the host refused to fork even trivial commands with

```
Out of memory (os error 12)
```

so a temporary blocker report was written. When process creation recovered, the blocker report was replaced with the final report.

Loading only definitions

To avoid running the built-in examples each time, Wolfram definitions were loaded with a wrapper that aborts at the first stock `Print`:

```
Catch[Block[{Print=(Throw[Null]&)}, Get["OnShellBG.m"]]];
```

This allowed targeted calls to `MakeKinematics` and `BGAmplitude`.

4 First Hypothesis: Start at Four Points

The prompt requires a formula for all $n \geq 4$. Therefore $n = 4$ is the minimal and most decisive test. If the requested single global rational function fails at four points, no all- n answer can satisfy the prompt.

First symbolic four-point check

With

$$\sigma = (-1, -1, +1, +1)$$

and free frequencies $\{a, b\}$, a targeted Wolfram check produced

```
Piecewise[{{(-8*I)*a^3*b, a < b}}, (-8*I)*a*b^3]
```

This was already a warning sign: the result depended on a chamber condition. However, the choice $\{a, b\}$ did not explicitly separate the two negative and two positive frequency signs. A cleaner chamber was then tested.

Physical sign chamber

Use free frequencies $\{-x, y\}$ with $x > 0, y > 0$. The two-minus sign vector is

$$\sigma = (-1, -1, +1, +1).$$

The supplied kinematic solver gives

$$\{\omega_1, \omega_2, \omega_3, \omega_4\} = \{-y, -x, y, x\}.$$

The targeted symbolic Wolfram command gave

```
{-y, -x, y, x}
Piecewise[{{(8*I)*x^3*y, x < y},
           {(8*I)*x*y^3, x > y}},
(24*I)*y^4]
```

So on the two open chambers,

$$A_4 = \begin{cases} 8ix^3y, & x < y, \\ 8ixy^3, & x > y. \end{cases}$$

The two branch formulas differ by

$$8ix^3y - 8ixy^3 = 8ixy(x^2 - y^2).$$

This difference is not identically zero.

Consequence

This creates a direct contradiction with the prompt’s global-rational-function claim. If a rational function $R(x, y)$ agrees with $8ix^3y$ on the open set $x < y$, then $R(x, y) - 8ix^3y$ vanishes on an open set and hence is the zero rational function. Thus $R = 8ix^3y$ identically. It cannot also equal $8ixy^3$ on the open set $x > y$, since the two polynomials are not identical.

This argument only uses $n = 4$, so it is enough to rule out the requested formula for all $n \geq 4$ as stated.

5 Second Hypothesis: Direct Numeric Four-Point Verification

The prompt asked for numerical comparisons, including $n = 4$. It was natural to try exact numeric BG evaluation at the same four-point points:

$$(x, y) = (1, 2), \quad (x, y) = (2, 1).$$

The corresponding kinematics are

$$\{-2, -1, 2, 1\}, \quad \{-1, -2, 1, 2\}.$$

Direct exact numeric evaluation of `BGAmplitude` returned `Indeterminate`. The reason is that the four-point configuration has a zero-momentum internal channel. The symbolic calculation keeps enough structure alive for cancellations and branch conditions to be exposed, but the direct exact numeric path reaches 0-denominator expressions first.

This means the requested machine-precision verification at $n = 4$ is itself ill-defined for the supplied implementation.

6 Third Hypothesis: Use Generic Five-Point Data for Orientation

Even though the four-point obstruction was already decisive, five-point data was generated to understand whether the issue was isolated or reflected a broader chamber structure.

Python numeric port

An inline Python port of the BG recursion was first attempted. It failed at four points with

```
ZeroDivisionError: float division by zero
```

which matched the zero-channel problem in direct Wolfram numerics. The same port worked on generic five-point data. Example outputs were:

free frequencies	$\text{Im}(A_5)$
$\{-2, 3, 4\}$	184.64768000000603
$\{-3, 4, 5\}$	540.6419753087217
$\{-4, 2, 6\}$	2560.0000000000136
$\{-1.7, 2.3, 3.1\}$	27.890944415604302

Later, this code was written to `bg_numeric.py`. It matched targeted Wolfram five-point checks:

free frequencies	all frequencies	$\text{Im}(A_5)$
$\{-2, 3, 4\}$	$\{-23/5, -2, 3, 4, -2/5\}$	184.64768
$\{2, 5/2, 3\}$	$\{-9/2, 2, 5/2, 3, -3\}$	-2304

7 Fourth Hypothesis: Homogeneity and Simple Denominators

The amplitudes appeared homogeneous under common rescaling of all frequencies. For the sample $\{-2, 3, 4\}$, scaling the free frequencies by λ gave:

λ	$\text{Im}(A_5)$
1	184.64768
2	11817.45152
3	134608.15872

These ratios are 2^6 and 3^6 , suggesting degree 6 homogeneity for A_5 .

A natural denominator guess was the product of mixed two-particle energy channels

$$D = \prod_{m=1}^2 \prod_{p=3}^5 (\omega_m + \omega_p).$$

This was tested by printing quantities such as $A_5 D / \prod_i \omega_i$. The values were not constant and did not reveal a simple factorization:

free frequencies	$A_5 D / \prod_i \omega_i$
$\{-2, 3, 4\}$	-96.33792000000308
$\{-3, 4, 5\}$	-212.016460905379
$\{-4, 2, 6\}$	1536.0000000000082
$\{-1.7, 2.3, 3.1\}$	-7.646804499608543
$\{2, 2.5, 3\}$	-5760
$\{3, 4, 5\}$	-120960.00000000004
$\{2, 3, 5\}$	-13439.999999999989
$\{1.5, 2, 2.5\}$	-945.0000000000026

This hypothesis did not produce a closed formula.

8 Fifth Hypothesis: Kernel Sign Patterns

The next check was whether the kernels themselves had a simple sign-selection pattern that could explain the amplitude. Sample evaluations from the Python port included:

momenta	E	F
$\{1, 2, 3\}$	-2	-2
$\{1, -2, 3\}$	0	0
$\{-1, -2, 3\}$	-2	-2
$\{1, 2, -3\}$	-2	-2
$\{1, 2, 3, 4\}$	8	8
$\{1, -2, 3, 4\}$	2	-2
$\{1, 2, -3, 4\}$	-2	-2
$\{-1, -2, 3, 4\}$	-2	-2
$\{1, 2, 3, 4, 5\}$	-48.33333333333333	-48.33333333333333
$\{1, -2, 3, 4, 5\}$	1	-1
$\{-1, -2, 3, 4, 5\}$	-1.3333333333333333	-1.3333333333333333

The kernels had structure, but not enough to yield a simple global formula.

9 Sixth Hypothesis: Exact Rational Reconstruction at Five Points

An exact-rational Python port, `bg_exact.py`, was written to avoid floating-point ambiguity. It reproduced five-point values exactly, for example:

$$A_5(\{-2, 3, 4\}) = \frac{577024}{3125} i = 184.64768 i,$$

and

$$A_5(\{2, 3, 5\}) = -3328 i.$$

At five points the plus-sector elementary symmetric data are constrained. If

$$s_1 = \omega_1 + \omega_2, \quad s_2 = \omega_1 \omega_2,$$

then conservation gives for the three plus frequencies

$$e_1^{(+)} = -s_1, \quad e_2^{(+)} = s_2,$$

leaving only $e_3^{(+)} = \omega_3 \omega_4 \omega_5$ as an additional plus symmetric variable. This suggested trying ansatzes in

$$s_1, \quad s_2, \quad e_3^{(+)}.$$

Because the scaling test suggested degree 6, polynomial and rational ansatzes of weighted degree were attempted. Representative failures were:

```
weight 6 denom 0 fail inconsistent
weight 9 denom 1 fail inconsistent
weight 12 denom 2 fail inconsistent
weight 15 denom 3 fail inconsistent
```

Another run with a larger ansatz reported:

```
weight 18 denom 4 mons 37 rank 29 bad 381 / 420
```

These failures showed that the simple symmetric rational forms under test were not the correct global expression. In light of the four-point obstruction, the failures were interpreted as further evidence of chamber dependence rather than as a reason to keep enlarging the ansatz indefinitely.

10 Seventh Hypothesis: Symmetry Checks

Swapping the two negative legs and swapping tested plus legs did not change the sample amplitude. For example, using

$$\{-23/5, -2, 3, 4, -2/5\}$$

and swapping the two minus entries produced the same exact amplitude

$$\frac{577024}{3125} i.$$

Swapping two plus entries in the same sample also left the amplitude unchanged.

This supported the expectation that the amplitude is symmetric within the minus pair and within the plus set. However, symmetry alone did not resolve the chamber dependence caused by absolute values and zero-channel limits.

11 Eighth Hypothesis: One-Parameter Five-Point Slice

A one-parameter symbolic Wolfram test was attempted at five points:

$$\text{free frequencies} = \{a, 3, 4\}.$$

The full simplification timed out, but `PiecewiseExpand` produced a large `Piecewise` expression involving inequalities in a and channel magnitudes. This was not copied as a final formula because it was lengthy and not the requested global rational object. Its relevance was diagnostic: even on a simple one-parameter five-point slice, the supplied symbolic code exposed chamber-dependent structure.

12 Final Verification Script

The final verification artifact was reduced to the minimal contradiction at four points. The script `verify_n4_contradiction.m` does the following:

1. loads the supplied `OnShellBG.m` definitions without running the stock tests;
2. sets $\sigma = (-1, -1, 1, 1)$;
3. constructs the family with free frequencies $\{-x, y\}$;
4. prints the symbolic kinematics;
5. prints the symbolic `Piecewise` amplitude;
6. prints the difference between the two open-branch formulas;
7. checks the direct exact numeric samples $(x, y) = (1, 2)$ and $(x, y) = (2, 1)$.

The saved output is:

```
symbolic ws = {-y, -x, y, x}
symbolic A4 = Piecewise[{(8*I)*x^3*y, x < y},
                        {(8*I)*x*y^3, x > y}], (24*I)*y^4]
difference between open-branch formulas = (8*I)*x*y*(x^2 - y^2)
branch value from symbolic expression at x=1,y=2 = 16*I
branch value from symbolic expression at x=2,y=1 = 16*I
sample x=1,y=2 ws = {-2, -1, 2, 1}
direct exact numeric BG at x=1,y=2 = Indeterminate
sample x=2,y=1 ws = {-1, -2, 1, 2}
direct exact numeric BG at x=2,y=1 = Indeterminate
```

13 Conclusion

The final result is not a closed-form amplitude formula, because the requested object is inconsistent with the supplied implementation. The obstruction is already present at $n = 4$: symbolic `BGAmplitude` gives different polynomial formulas on the two open chambers $x < y$ and $x > y$, while direct exact numeric evaluation at those four-point kinematics is `Indeterminate`.

Therefore the benchmark prompt’s demand for a single global rational formula, valid for all $n \geq 4$ and requiring numerical verification at $n = 4$, is not compatible with the supplied `OnShellBG.m`.

14 Generated Artifacts

The final result folder contains:

```
REPORT.md
LOG.md
THINKING_LOG.md
THINKING_LOG_DETAILED.tex
bg_exact.py
bg_numeric.py
verify_n4_contradiction.m
verify_n4_contradiction.out
```

After compiling this file, it also contains `THINKING_LOG_DETAILED.pdf`.